

A photonic integrated mode-locked laser based on dispersion-managed mode-locking architecture

Xurong Li,^{1,2} Zheru Qiu,^{1,2} Xuan Yang,^{1,2} Xinru Ji,^{1,2} Jiale Sun,^{1,2} Jianqi Hu,^{1,2}
Grigorii Likhachev,^{1,2} Ulrich Kentsch,³ and Tobias J. Kippenberg^{1,2,*}

¹ Institute of Physics, Swiss Federal Institute of Technology Lausanne (EPFL), CH-1015 Lausanne, Switzerland

² Institute of Electrical and Micro Engineering, Swiss Federal Institute of Technology Lausanne (EPFL),
CH-1015 Lausanne, Switzerland

³ Helmholtz-Zentrum Dresden-Rossendorf (HZDR), 01328 Dresden, Germany

*tobias.kippenberg@epfl.ch

Abstract: We demonstrate the first self-starting, photonic integrated femtosecond mode-locked laser, using a dispersion-managed mode-locking architecture. The laser has a 1.2-GHz repetition rate with an ultra-low mode-locking threshold, and generates a passively stable optical frequency comb. © 2025 The Author(s)

1. Introduction

Femtosecond mode-locked lasers (MLLs) are essential tools in numerous areas, from high-precision optical frequency metrology, tracking chemical reaction dynamics, time distribution, to corneal surgery [1]. While solid-state and fiber MLLs still offer the best performance, miniaturized femtosecond MLLs are attractive for their better size, weight, and power (SWaP) as well as lower cost. However, current integrated ultrafast sources still fall significantly short of the performance compared to the bench-top counterparts. Kerr soliton microcombs have repetition rates from tens of GHz to THz range and usually have low conversion efficiency [2]. Integrated electro-optic comb generators can produce broadband spectra, but require external microwave sources [2]. Active MLLs on thin-film lithium niobate have been demonstrated, while producing only picosecond pulses and requiring external microwave driving [3]. Integrated semiconductor MLLs typically only generate picosecond pulses with poor noise performance [4]. Rare-earth-doped waveguide MLLs have so far been limited to pulse energies around 5 pJ with a relatively large footprint in planar lightwave circuits [5]. Recent efforts to integrate these systems in photonic integrated circuits (PICs) have only yield Q-switching [6] or Q-switched mode-locking [7] operations. The high nonlinearity associated with high-energy optical pulses confined in nano-scale waveguides can lead to wave breaking [8] and make passive mode-locking in PICs challenging.

Here, we demonstrate the first turnkey photonic integrated passive MLL, based on a dispersion-managed mode-locking architecture. The MLL consists of three main components: erbium-doped silicon nitride (Er:Si₃N₄) waveguides, chirped Bragg gratings (CBGs), and a semiconductor saturable absorber mirror (SESAM). It can reach stable mode-locking state at a pump power of only 27.3 mW, and generate optical pulses with repetition rate of 1.2 GHz and pulse duration of 300 fs. The ultra-low mode-locking threshold enables turnkey operation, deterministic multi-soliton generation, and battery operation of a compact MLL system. It provides exceptional comb-line stability, two-order-of-magnitude better than that of a commercial fiber MLL, and superior noise performance, three-order-of-magnitude better than that of the best-reported integrated MLL [9].

2. Dispersion-managed mode locking

In dispersion-managed mode-locking architecture, normally-dispersive and anomalously-dispersive sections are distributed along the cavity, resulting optical pulses to alternately stretch and compress as they circulate inside the cavity and reducing the averaged peak power and nonlinearity. In our MLL, the normal-dispersion Er:Si₃N₄ waveguides, which are more immune to pulse breaking [8], provide optical gain. The CBGs simultaneously serve as a cavity mirror, an output coupler, a pump injector, and a dispersion compensator, offering net-cavity anomalous dispersion to support dispersion-managed soliton [10]. A SESAM, which is commercially available, initiates mode-locking by suppressing low-intensity noises and reflecting high-intensity pulses.

Our MLLs are fabricated on a 200-nm Si₃N₄ PIC platform on 4-inch silicon wafers with wafer-scale erbium ion implantation. The MLL chip only occupies a 5 × 5-mm² area with 12 individual MLLs and ~ 7-cm cavity lengths. CBGs have 2000-3000 grating periods, linearly chirped to cover Bragg wavelengths from 1500 to 1600

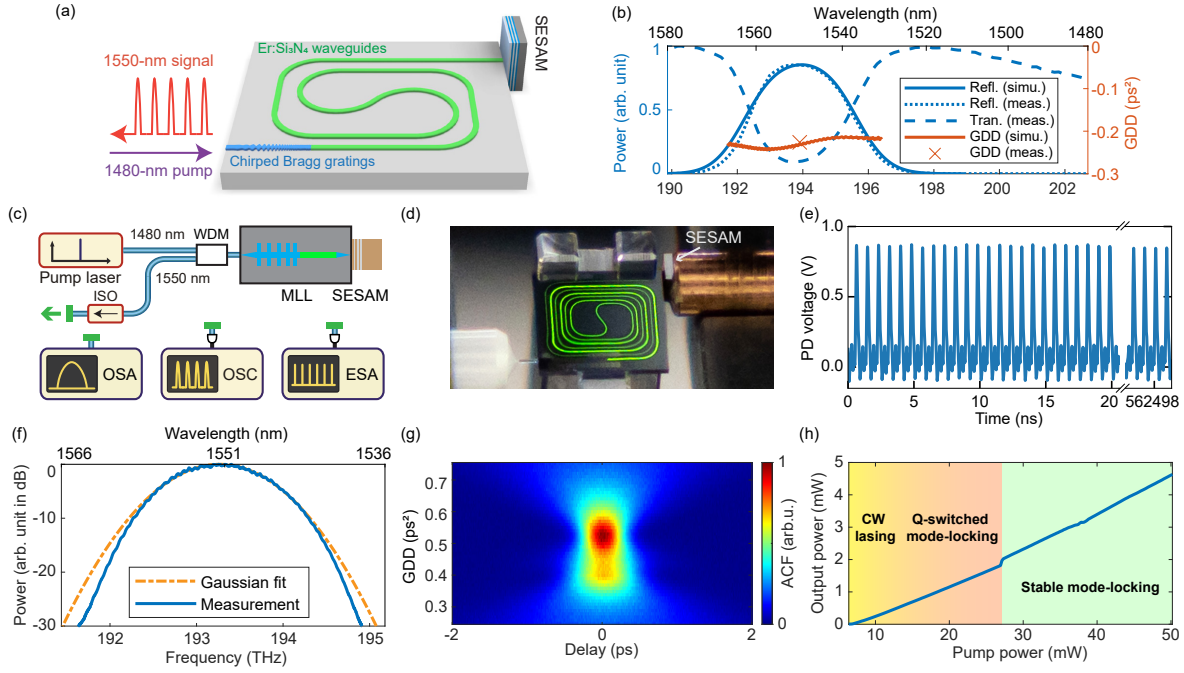


Fig. 1. Photonic integrated dispersion-managed mode-locked lasers. (a) Schematic of the photonic integrated MLL. (b) Simulation and measurement results of the chirped Bragg gratings' reflection, transmission, and group-delay dispersion (GDD). Reflection and transmission spectra are normalized to transmission's maximum response. (c) Schematic diagram of the experimental setup for MLL's operation and characterization. OSC: oscilloscope. OSA: optical spectrum analyzer. ESA: electrical spectrum analyzer. (d) A photograph of a photonic-integrated MLL during operation. (e) Photodetected stable pulse train. (f) MLL's output optical spectrum in logarithmic scale. (g) Measured autocorrelation function (ACF) as a function of GDD provided by a pulse shaper. (h) On-chip output power of an ultra-low-threshold MLL as a function of the on-chip pump power.

nm. Measured CBG characteristics match well with the design values (Fig. 1b), with a 26.1-nm 3-dB reflection bandwidth centered around 1550 nm, high transmission at the 1480-nm pump wavelength, and a group-delay dispersion (GDD) of -0.226 ps^2 .

The operation of MLL requires only 5 components: the MLL chip, a SESAM, a pump laser, a wavelength-division multiplexer (WDM), and an isolator (Fig. 1c). The MLL can generate stable pulse train at a 1.2-GHz repetition rate (Fig. 1e), with on-chip output power up to 22.5 mW and the 3-dB bandwidth up to 9.5 nm (Fig. 1f). Since the output optical pulses are highly chirped, we used a pulse shaper and an autocorrelator to mimic the linear pulse compression in fiber and measure the transform-limited pulse duration. The minimum full-width-half-maximum is 424.7 fs, corresponding to a 300-fs pulse width for a Gaussian-type pulse (Fig. 1g). The MLL can reach stable mode-locking state at only 27.3-mW pump power by incorporating pump-recycling gratings at the end of the Er:Si₃N₄ waveguide before the SESAM port. The ultra-low mode-locking threshold enables a compact MLL system powered by only 4 AA batteries.

3. Passively stable optical frequency comb

The regular pulse train generated by our photonic integrated MLL (PIC-MLL) corresponds to an optical frequency comb in the spectral domain. To validate its comb nature, we heterodyne-beat our MLL with a frequency-stabilized reference laser, and compare it with a commercial fiber MLL (Menlo ELMO) in the same experiment (Fig. 2a,b). While the comb line of commercial fiber MLL drifts more than 33 repetition rates within 1 hour, our PIC-MLL exhibits exceptional comb-line stability: the standard deviation of the comb-line drift during the 2-hour measurement is only 8.0 MHz, corresponding to less than 1% of the repetition rate. Allan deviations of the comb-line drift and comb-line-number-scaled repetition rate further suggest that the PIC-MLL has two-order-of-magnitude higher comb-line stability than the commercial fiber MLL at an averaging time of 1000 s (Fig. 2c).

The MLL also exhibits narrow optical linewidths with high uniformity. We measured the frequency noise of the beatnote at 6 different wavelengths across 1544 comb lines, and the variations are only 8.6% and 12.3%

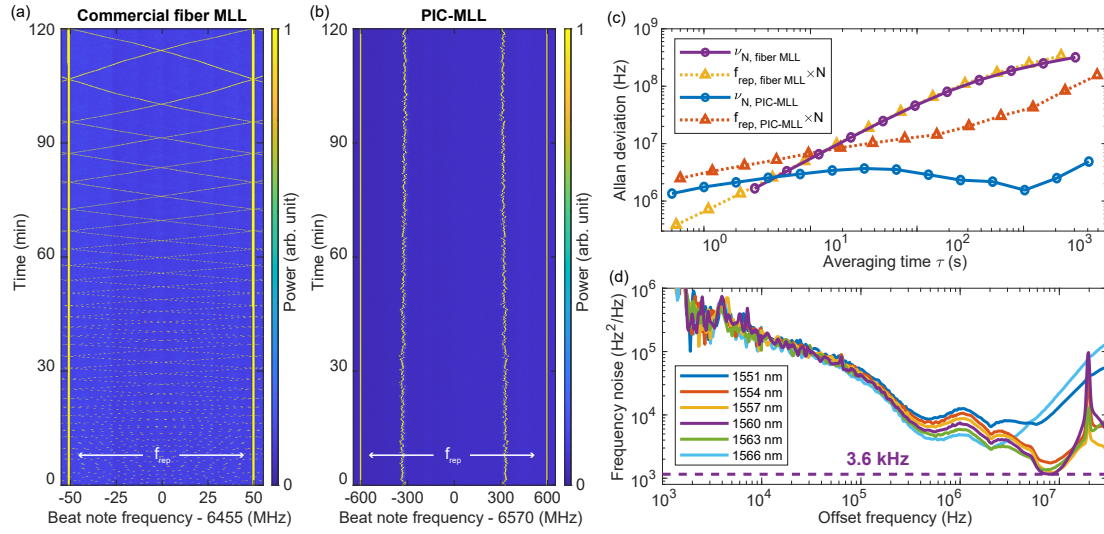


Fig. 2. **A passively stable optical frequency comb.** (a,b) Spectrogram of heterodyne beat notes of a commercial fiber MLL (a) or a PIC-MLL (b) with a frequency-stabilized reference laser at 1559.424 nm over 2 hours. (c) Allan deviation of the PIC-MLL and commercial fiber MLL's comb-line drift and comb-line-number-scaled repetition rate. (d) PIC-MLL's frequency noise of heterodyne beat notes at different comb lines, with the minimum linewidth of 3.6 kHz.

at 10-kHz and 100-kHz, respectively. The fitted minimum Lorentzian linewidth is 3.6 kHz, more than 40 times narrower than that of the best reported integrated MLL [9]. We also measured the single-sideband phase noise of the photodetected repetition-rate signal, showing a timing jitter of 46.9 fs integrated from 10 kHz to 10 MHz. The extracted Lorentzian linewidth of the repetition-rate signal is 0.4 mHz, three-order-of-magnitude better than that of the best reported integrated MLL [9].

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